

TEXTBOOK CHAPTER LECTURE

Interaction Design: Beyond Human-Computer Interaction

Helen Sharp, Yvonne Rogers, Jennifer Preece — 5th Edition, Wiley

Chapter 1

What Is Interaction Design?

*Good & Poor Design • Interdisciplinary Foundations • The User Experience
Understanding Users • Accessibility • Usability & UX Goals • Design Principles*

By the End of This Chapter, You Will Be Able To



Explain the difference between good and poor design using everyday examples



Define interaction design and describe its interdisciplinary foundations



Describe what is meant by “the user experience”



Explain why understanding users is central to the design process



Discuss accessibility and inclusiveness in interactive systems



Identify usability goals and user experience goals



Apply core design principles: visibility, feedback, constraints, consistency, affordance

Chapter 1 Structure

1.1 Introduction

1.3 What Is Interaction Design?

1.5 Understanding Users

1.7 Usability and User Experience Goals

1.2 Good and Poor Design

1.4 The User Experience

1.6 Accessibility and Inclusiveness

1.8 Design Principles

Why Study Interaction Design?

Technology is now woven into almost every part of daily life — phones, cars, appliances, websites, and workplace software all depend on how well people can interact with them.



When interaction fails

Users make errors, feel frustrated, or abandon a product entirely



When interaction succeeds

Products feel invisible — people accomplish goals with little conscious effort



Chapter 1's purpose

Introduces the vocabulary and mindset used throughout the rest of the book

The Classic Example: Doors



The “Norman Door” Problem

- A flat door plate on both sides gives no clue whether to push or pull
- Users hesitate, push the wrong side, or pull when they should push
- The problem is not the user — it is a design that fails to communicate its own operation



A Well-Designed Door

- A flat plate signals “push”; a handle or bar signals “pull”
- The design itself communicates the correct action — no sign or instruction needed
- This mapping between appearance and action is called an affordance

Poor Design in Everyday Technology



Overloaded remote controls

Dozens of unlabeled, similarly-sized buttons force users to guess or hunt for the right one



Cluttered software menus

Too many options with no clear hierarchy overwhelm first-time users



Confusing elevator panels

Unclear floor numbering and button layout leads people to press the wrong floor



Inconsistent dashboard icons

Symbols that don't match users' expectations slow down recognition while driving

What Separates Good Design from Poor Design?

Across all these examples, good design consistently does four things:



Makes the right action obvious

Through visible cues (affordances) rather than instructions



Gives immediate feedback

So users always know whether their action succeeded



Constrains choices sensibly

Reducing the chance of pressing the wrong control



Behaves consistently

So knowledge from one part of the system transfers to another

These four ideas reappear later in this chapter as formal design principles.

1.3 WHAT IS INTERACTION DESIGN?

Defining Interaction Design

“Designing interactive products to support the way people communicate and interact in their everyday and working lives.”

— Sharp, Rogers & Preece

In practice, this means designers must:



Involve users

Throughout design, not just at the end



Balance trade-offs

Between competing needs, constraints, and goals



Iterate

Test, learn, and refine designs repeatedly

1.3 WHAT IS INTERACTION DESIGN?

An Interdisciplinary Field

Interaction design draws on knowledge and methods from many disciplines:



Psychology

Understanding perception, cognition & behaviour



Computer Science

Building the underlying software & systems



Sociology & Anthropology

Studying people in social & cultural context



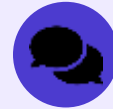
Ergonomics

Fitting systems to human physical capabilities



Graphic & Product Design

Crafting visual form and physical shape

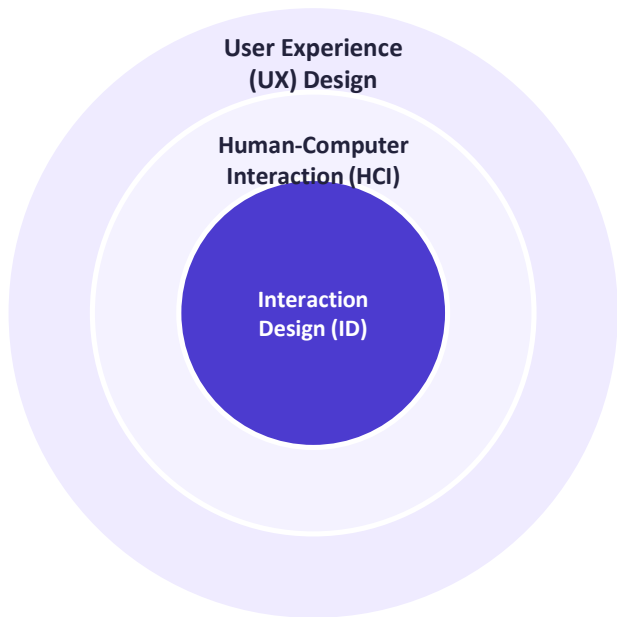


Linguistics

Structuring clear, understandable communication

1.3 WHAT IS INTERACTION DESIGN?

How Interaction Design Relates to HCI and UX



UX Design

The broadest scope — covers the entire experience, including branding, usability, and function



HCI

The academic field studying how humans and computers interact, going back to the 1980s



Interaction Design

A core practice within HCI/UX focused specifically on designing the interaction itself

1.3 WHAT IS INTERACTION DESIGN?

Who Practises Interaction Design?

Real products are rarely designed by one person — they emerge from a team of specialists:



UX Researcher

Studies user needs, behaviours, and pain points



Interaction Designer

Designs how users accomplish tasks step by step



Visual / UI Designer

Shapes the look, feel, and visual hierarchy



Information Architect

Organizes content and navigation structure



Front-End Developer

Implements the interface in working code



Usability Analyst

Tests designs against usability & UX goals

What Do We Mean by “User Experience”?

Every aspect of how a person feels about using a product — how it makes them feel, whether it is satisfying, and how effectively it fits into their life.

- UX cannot be designed directly — designers can only design for a user experience, since the experience itself is subjective and personal
- It goes beyond usability: a system can be perfectly usable yet feel cold, boring, or untrustworthy
- Classic examples: the original iPod's simplicity, or TiVo's remote redesigned around how people actually watch television

Factors That Shape the User Experience



Emotional response

Delight, frustration, trust, or anxiety while using the product



Aesthetics

Visual design, style, and overall polish



Motivation

Whether the product encourages continued, willing use



Context of use

Physical setting, social situation, and available time



Prior experience

Expectations shaped by similar products used before



Time pressure

How rushed or relaxed the user is during the task

Desirable vs. Undesirable UX Qualities

When evaluating a design, it helps to name the quality you want it to have — or avoid:

Desirable

- ✓ Satisfying
- ✓ Enjoyable / fun
- ✓ Engaging
- ✓ Pleasurable
- ✓ Exciting
- ✓ Helpful
- ✓ Motivating
- ✓ Rewarding
- ✓ Emotionally fulfilling
- ✓ Supports creativity

Undesirable

- X Boring
- X Frustrating
- X Annoying
- X Patronizing
- X Making one feel guilty
- X Making one feel stupid
- X Cutesy / gimmicky
- X Unpleasant
- X Confusing
- X Alienating

1.5 UNDERSTANDING USERS

Users Are Not All the Same

Designing well requires recognizing how people differ along several dimensions:



Physical & sensory abilities

Age, dexterity, vision, hearing, and motor control vary widely



Cognitive characteristics

Memory, attention, prior knowledge, and learning speed differ



Social & cultural background

Language, norms, and familiarity with technology shape expectations



Role & expertise

A novice, an occasional user, and an expert need different support

Designing for Real People: Personas



What Is a Persona?

A fictional but realistic profile representing a key group of users, built from real research data — not guesswork.

- Name, goals, and frustrations
- Typical tasks and context of use
- Level of experience with technology



Why Personas Help

- Keep the whole team focused on real needs, not personal preference
- Prevent designing only for “users like me”
- Provide a shared reference point during design debates

Accessibility



The extent to which a product can be used by people with disabilities — visual, auditory, motor, speech, or cognitive.



Visual

Screen readers, high contrast, resizable text



Hearing

Captions, visual alerts, transcripts



Motor

Keyboard navigation, larger touch targets



Cognitive

Simple language, consistent layout, clear steps

Inclusiveness and Universal Design

Inclusiveness means designing so the widest possible range of people can use a product effectively — across age, literacy, culture, income, and language.



The Curb-Cut Effect

Sidewalk ramps were built for wheelchair users — but today they also help parents with strollers, travellers with luggage, and delivery workers with carts. Designing for accessibility often improves the experience for everyone, not just the original target group.

Usability Goals — Concrete & Measurable



Effective

Does it help users achieve what they set out to do?



Efficient

Can tasks be completed quickly with minimal effort?



Safe

Does it prevent errors, or make recovery easy?



Good utility

Does it provide the right functions for the task?



Easy to learn

Can new users become competent quickly?



Easy to remember

Can occasional users recall how it works?

1.7 USABILITY AND UX GOALS

User Experience Goals — Subjective & Hedonic

These describe how a product should feel, not just how it should function.



Satisfying



Fun / enjoyable



Motivating



Aesthetically pleasing



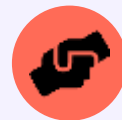
Supports creativity



Rewarding



Emotionally fulfilling



Helpful & supportive

Five Core Design Principles

Generalizable rules — distilled from usability and UX goals — that guide everyday design decisions:



Visibility

Relevant controls and information should be visible to the user



Feedback

Every action should produce a clear, immediate response



Constraints

Limiting possible actions reduces the chance of error



Consistency

Similar operations and elements should behave the same way



Affordance

An object's appearance should suggest how it is used

Principle: Visibility



The more visible a function is, the more likely users will know it exists and how to use it.

- Hidden or buried functions are functions users never discover
- Example: a clearly visible “Save” icon vs. a save function buried three menus deep
- Physical controls (buttons, switches) are inherently more visible than gestures with no visual cue

Principle: Feedback



Sending information back to the user about what action has been done and what has been accomplished.

- Can combine sound, visual highlighting, animation, and text messages
- Example: a button that changes colour when pressed, or a progress bar during upload
- Without feedback, users repeat actions unnecessarily or assume the system has failed

Principle: Constraints



Restricting the possible actions that can be performed, guiding the user toward the correct one.



Physical

Shape or size restricts how an object can be handled (e.g., a plug that only fits one way)



Logical

The order of steps determines what makes sense to do next



Cultural

Established conventions guide expectations (e.g., a red icon means stop or delete)

Principle: Consistency



Designing interfaces so similar operations and elements are used or invoked in the same way, every time.

- Once a user learns one part of a consistent system, that knowledge transfers elsewhere
- Example: if swiping left deletes an item in one screen, it should do the same on every screen
- Inconsistency forces users to relearn behaviour and increases the chance of error

Principle: Affordance



A property of an object that suggests how it should be used — a handle affords pulling, a button affords pushing.



Perceived affordance

What the interface visually suggests it can do (works well for screen-based UI)



Real affordance

What a physical object actually allows (works well for physical products)

SUMMARY

Chapter 1 in Six Ideas

- 1 Good design communicates its own use; poor design forces guesswork
- 2 Interaction design is interdisciplinary — psychology, computer science, sociology, and design all contribute
- 3 The user experience covers feelings, not just functionality
- 4 Users differ physically, cognitively, and culturally — design for the range, not the average
- 5 Accessible, inclusive design tends to benefit everyone, not only its original target group
- 6 Usability goals + UX goals + design principles together define what “good” means in practice

Key Terms to Know

Interaction design

Designing interactive products to support how people communicate and work

Usability goal

A concrete, measurable target such as efficiency or safety

Affordance

A property suggesting how an object should be used

User experience (UX)

How a product makes a person feel, beyond pure functionality

UX goal

A subjective quality such as being satisfying or motivating

Persona

A realistic profile representing a key user group

Primary Reference



Sharp, H., Rogers, Y., & Preece, J.

“Interaction Design: Beyond Human-Computer Interaction,” 5th Edition, Wiley, 2019 — Chapter 1.

Related Reference Books (from course syllabus)

- Gerard Jounghyun Kim, “Human Computer Interaction — Fundamentals and Practice,” CRC Press, 2015.
- Ben Shneiderman et al., “Designing the User Interface,” 5th Edition, Pearson, 2009.
- Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, “Human-Computer Interaction,” 3rd Edition, Pearson, 2003.